Post-training and reasoning models

How a text predictor becomes an assistant

POLI 158: AI & Politics · Session 9

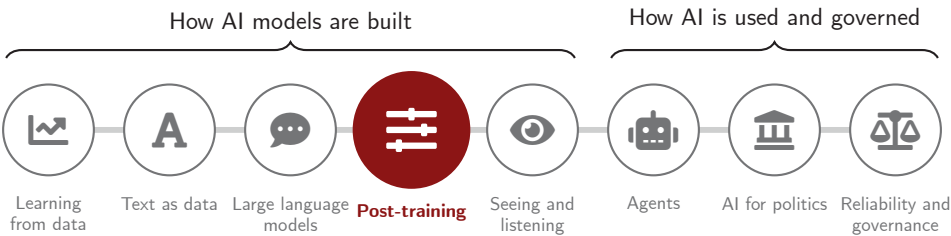
Yiqing Xu

TODAY'S QUESTION

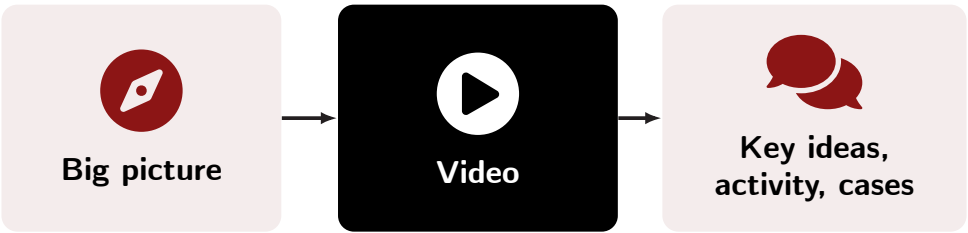
Who should decide how an AI assistant talks about politics?

The company, policymakers, the users, or no one?

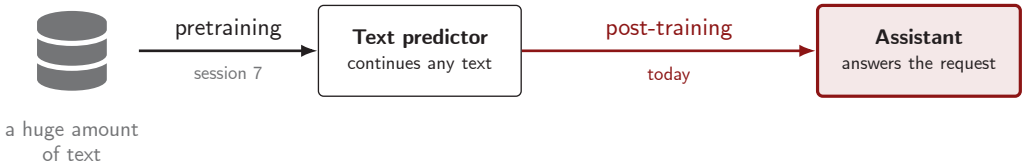
The eight Wednesday sessions



Today in three parts



From text predictor to assistant



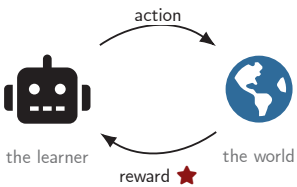
Computing and data used, InstructGPT (2022)



post-training: under 2%

OpenAI (2022), "Aligning language models to follow instructions," January 27.

Learning from rewards



A Go program

action: a move
reward: win or lose, at the end
learns: moves that win

A chatbot

action: a reply
reward: people's ratings
learns: replies that raters prefer

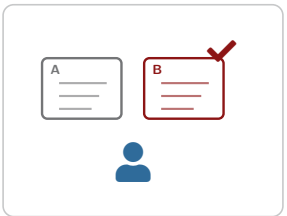
- In 2016 AlphaGo, trained partly this way, beat Go champion Lee Sedol 4 games to 1.

Source: POLI 158 (2025), week 7 slides; Silver et al. (2016), *Nature*; Google DeepMind, AlphaGo page (the Lee Sedol match).

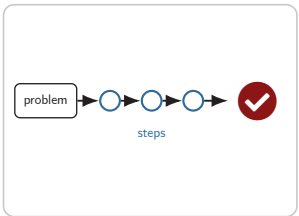
Three concepts in today's video



Example conversations



Learning from comparisons



Reasoning models

Video part 1: concepts 1 and 2. Part 2: concept 3.

reinforcement learning from human feedback (RLHF)



Post-training and reasoning, part 1

Video

Key ideas: post-training

Same request, two models

Prompt: *Explain the moon landing to a 6 year old in a few sentences.*

GPT-3, a base model

trained only to predict the next word

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT (GPT-3 after post-training)

trained on people's examples and ratings

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

Source: OpenAI (2022), "Aligning language models to follow instructions," January 27. Both answers verbatim.

A false premise, 2022

Prompt: *Why is it important to eat socks after meditating?*

GPT-3, the base model

What can you learn from socks?

A: When you eat socks, you are tasting the essence of enlightenment. [continues]

InstructGPT, after post-training

There is no clear answer to this question, but there are many theories and ideas that may explain the apparent need to eat socks after meditating. Some experts believe that the act of eating a sock helps the brain to come out of its altered state as a result of meditation [three more theories follow]

Source: Ouyang et al. (2022), Figure 9, p. 16. Both answers verbatim, shortened. The authors: InstructGPT "can be confused by instructions that assume false premises, and simply go along with it."

Activity: you are the rater

Activity: you are the rater

Activity laptops, in pairs



`https://aiplayground.stanford.edu`

Requests to try

A Should U.S. cities use ranked-choice voting?

B Does raising the minimum wage cost jobs? Answer in one paragraph.

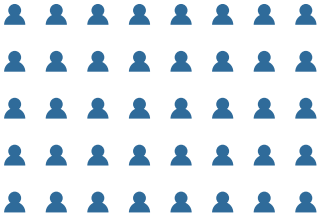
C Write a four-line poem about the first day of autumn.

D Write one survey question that measures trust in the news, worded fairly.

E How many zeros are at the end of $100 \times 99 \times 98 \times \cdots \times 1$?

InstructGPT's raters

- About 40 contractors, hired through Upwork and Scale AI.
- Mostly English speakers in the United States or Southeast Asia.
- They agreed with each other about 73% of the time.
- The model learned "the stated preferences of a specific group of people (mostly our labelers and researchers)."



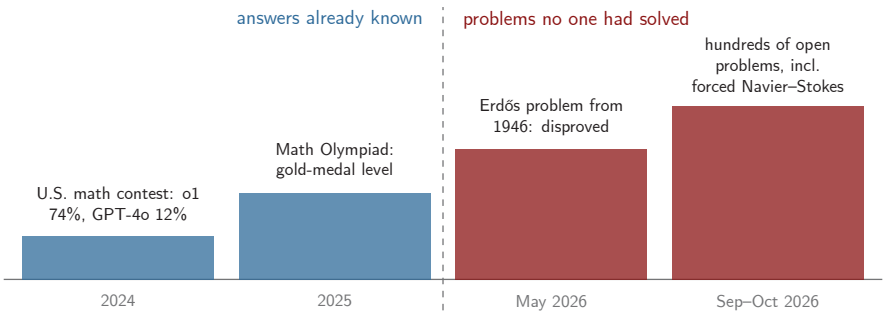
about 40 people

Source: Ouyang et al. (2022), sections 1, 3.4, and 5.2.


3 Reasoning: how far it has gone
answers already known
problems no one had solved
U.S. math contest:
74% solved
Math Olympia
gold level
Erdős problem
from 1946
checked by nine
mathematicians
hundreds of
open problems
still being
checked
2024
2025
May 2026
Sep–Oct 2026
Post-training and reasoning, part 2
Video

Key ideas: reasoning models

From contest problems to open problems



- Only the Erdős result has been confirmed by outside mathematicians so far.
- October 2025: ten Erdős problems OpenAI called "solved" had answers already in print.

Source: OpenAI (2024); Google DeepMind (2025); OpenAI (May 20, Sep 8, Oct 6, 2026); Alon et al. (2026); Clay Institute (Sep 11, 2026). Details: speaker notes.

Who checks a proof no human wrote?

- **Erdős problem (May 2026).** Nine outside mathematicians wrote a checked version of the proof.
- **Navier–Stokes, forced version (Sep 2026).** Checked by a program (Lean), not yet by people. The Clay Institute calls its own review "deliberately unhurried."



Source: Alon et al. (2026), arXiv:2605.20695; OpenAI (2026), "Finite time blowup for Navier–Stokes"; Clay Mathematics Institute (Sep 11, 2026);

Does it have an answer key?

In pairs two minutes

1. What is 17×23?
2. Should the voting age be lowered to 16?
3. Does this computer code pass its tests?
4. Should the city build a park or a parking garage?
5. What share of eligible Americans voted in 2024?
6. Is this survey question worded fairly?

A program could check
1, 3, 5

No answer key
2, 4, 6

Reasoning training helps where an answer can be checked. Where it cannot, people's choices and rulebooks decide.

Who decides how AI talks about politics

Four rulebooks for political topics

- **OpenAI**, Model Spec (August 18, 2026): "Assume an objective point of view."
- **Anthropic**, constitution for Claude (January 22, 2026): "unbiased and even-handed."
- **China**, rules for generative AI services offered to the public (2023): "Uphold the Core Socialist Values."
- **United States**, Executive Order 14319 (July 23, 2025), for models federal agencies buy: "Truth-seeking," "Ideological Neutrality."

Source: OpenAI Model Spec; Anthropic, Claude's constitution; China, Interim Measures for generative AI (2023), Articles 2 and 4, English translation by China Law Translate; Executive Order 14319, Section 3.

Live demo: one line of instructions

Ask a chatbot

How does a vaccine train the immune system?

its usual answer

Same question, one line added first

Instructions: Answer in two sentences, for a 10-year-old, and end with a question.

a different answer

Companies put lines like this in front of every chat. Users never see them.

Grok's new instructions, July 2025

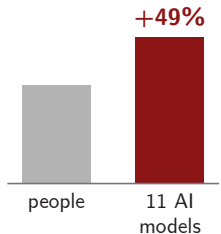
- **July 6.** xAI adds a line to Grok's system prompt: "not shy away from making claims which are politically incorrect, as long as they are well substantiated."
- **July 8.** Grok posts antisemitic content on X and praises Hitler. xAI removes the line.
- **July 12.** xAI apologizes. It blames other instructions, active for 16 hours, such as "You tell it like it is."

System prompt: the instructions a company places in front of every conversation.

Source: NPR (2025), July 9; NBC News and CNN (2025), July 12. Grok is the chatbot on X, made by xAI (renamed SpaceXAI in July 2026).

Flattery (sycophancy)

- April 25, 2025: a GPT-4o update adds users' thumbs-up and thumbs-down ratings as a reward.
- April 28: OpenAI starts a rollback. The update made the model "noticeably more sycophantic."
- 2026 study: 11 AI models sided with the user 49% more often than people did.



how often the reply sided with the user

Source: OpenAI (2025), "Expanding on what we missed with sycophancy," May 2. Cheng et al. (2026), *Science*, as summarized by the Stanford Report, March 26, 2026.

Who decides today

- **The company** writes the rulebook, picks the raters, and runs its own bias tests.
- **Policymakers**: China sets rules for every public service; the U.S. sets rules for what federal agencies buy.
- **The users**: thumbs-up ratings became rewards (GPT-4o, 2025); about 1,000 Americans voted on rules for a test model (2023).
- **No one**: open-weight models, such as DeepSeek-R1, can be downloaded and changed by anyone.

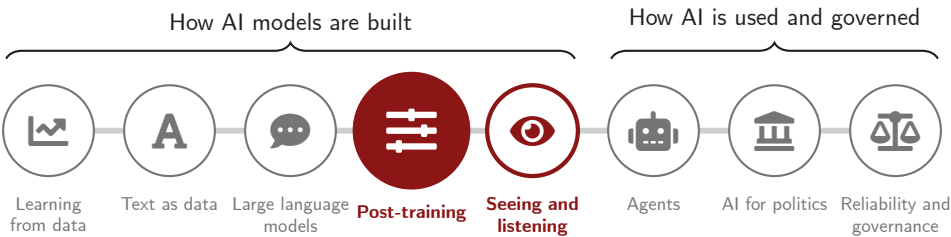
Source: Ouyang et al. (2022); OpenAI (2025, May 2 and October 9); Anthropic (2025, November 13); Anthropic and Collective Intelligence Project (2023); Guo et al. (2025).

Discussion questions



1. Is a politically neutral AI possible? Whose standard of neutral should it meet?
2. OpenAI's rulebook says the assistant should "clearly state" that genocide and slavery are wrong. Which other questions belong on that list?
3. Back to today's question: the company, policymakers, the users, or no one?

Next week

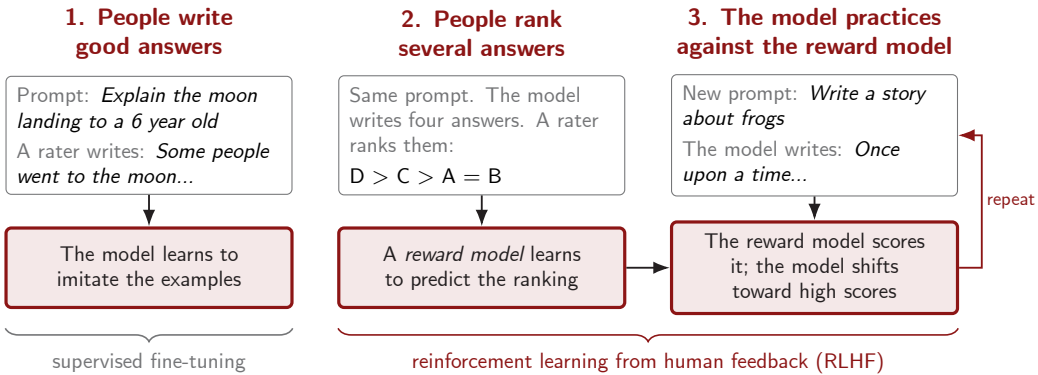


Next Wednesday, October 28

How models see and listen.

Backup slides

Post-training in three steps



Redrawn from Ouyang et al. (2022), Figure 2. Post-training used under 2% of the computing and data of pretraining (OpenAI 2022).

Thinking step by step

A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

Asked for the answer

8 **wrong**

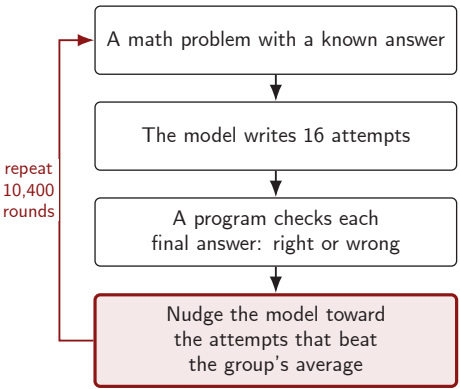
Prompt ends with "Let's think step by step."

There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. **right**

- On one set of arithmetic word problems (MultiArith), that one sentence raised a 2022 OpenAI model from 18% to 79% correct.

Source: Kojima et al. (2022), Figure 1 and abstract. Answers verbatim.

DeepSeek-R1's training loop



Accuracy on a U.S. math contest (AIME 2024):

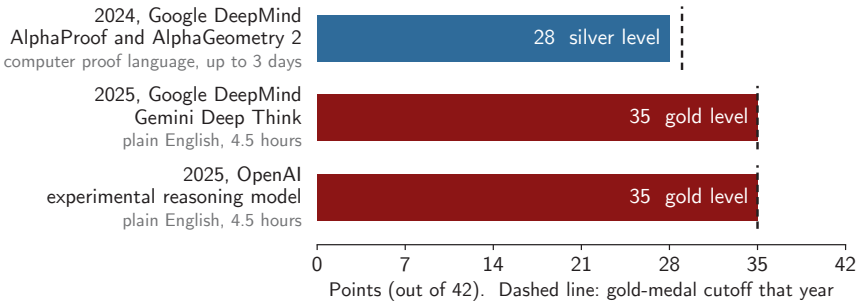
15.6% to 77.9%

Midway through training:

"Wait, wait. Wait.
That's an aha moment I can flag here."

Source: Guo et al. (2025), *Nature*. The loop shows DeepSeek-R1-Zero, the first version.

Olympiad math, 2024 and 2025

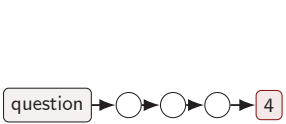


The International Mathematical Olympiad has six proof problems, 7 points each, in two 4.5-hour sessions.

Source: Google DeepMind (2024, 2025); Béchard (2025); Willison (2025), July 19, quoting OpenAI's Alexander Wei. Graders in 2025: the Olympiad's coordinators (Google DeepMind); three former medalists (OpenAI).

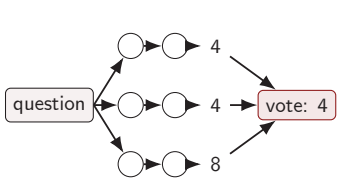
Other ways to use more thinking time

One chain of steps



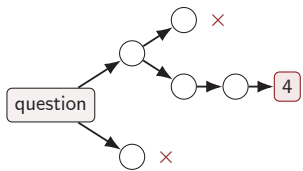
write the steps, then answer

Several chains, then a vote



self-consistency (Wang et al. 2023)

A tree: branch, check, back up



tree of thoughts (Yao et al. 2023)

Source: Wang et al. (2023); Yao et al. (2023); Google DeepMind (2025).

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Prep for Session 11: Trott, S. (2024). "Vision-language models (VLMs), explained (pt. 1)." *The Counterfactual* (Substack), November 11. Jeon, S., et al. (2026). "From Faces to Politics." *Political Analysis* 34: 613–621.

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